Project 10: AlphaFold understanding it through usage

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Abstract

AlphaFold has revolutionized structural biology, yet its predictive confidence and topological accuracy vary significantly across protein types and structures, particularly intrinsically disordered proteins (IDPs) and dynamic multi domain complexes. Building upon the Arity packing signatures introduced by Cazals and Sarti (2024), this study introduces a multiscale pipeline to evaluate 8,050 PDB and AF2 paired structures. By introducing a sequence gap parameter k to isolate long range tertiary packing from local secondary structure, alongside dynamic sliding window W filters. The findings statistically reveal AF2's systemic biases relative to PDB experimental data. Furthermore, Dynamic Time Warping DTW and confusion matrix analyses confirm that filtering trivial noise maximizes global topological alignment.

1 Introduction

While AlphaFold 2 has largely solved the protein folding problem, its performance across different protein types, like IDPs, molten and rigid globulars, remains heavily scrutinized. Recent evaluations, such as those by Cazals and Sarti, utilize C_α arity to bypass standard sequence alignment and create 2D topological signatures that distinguish dense folded regions from sparse and disordered sequences. This project replicates and expands upon the baseline Arity methodology by introducing a multiscale sequence gap $k > 0$ and a structural smoothing window W, where both of them mathematically isolate long range domain topology. Additionally, through Earth Mover's distance (EMD), Dynamic Time Warping (DTW) and Agglomerative Clustering, AF2 is systematically compared against the baseline Protein Data Bank to uncover structural and quaternary biases.

2 Theory & Methodology

The following pipeline follows a computational methodology with its respective theory separated in four phases and designed to evaluate the structural predictions of AlphaFold 2 against experimental reality.

2.1 Phase 1: Data Acquisition—Geometric Parsing—Synchronization

The objective of the first phase is to translate 3D Cartesian coordinates into perfectly aligned topological spatial Arity data, that resemble time series.

To begin with, a dataset of roughly 8,050 proteins that have experimentally resolved structures in the Protein Data Bank was downloaded alongside predictions from AlphaFold 2. Concurrently all auxiliary files required for the next phases were fetched including the AlphaFold Predicted Aligned Error PAE JSONs. These coordinate files were parsed to isolate the C_α atom coordinates and then mapped their experimental PDB sequence to the full AlphaFold sequence using the **Smith-Waterman** local alignment algorithm using a BLOSUM62 substitution matrix and wherever there are missing unresolved PDB indices, they are padded with null values to maintain sequential alignment.

Afterwards, both standard **local arity and long range arity** with $k > 0$ for each C_α atom are calculated. Let the arity of a C_α carbon be the number of neighboring C_α atoms within a distance

range r. Assuming the polypeptide chain has n amino acids the arities are defined as $L_a = \{a_1, ..., a_n\}$ of the n C_α carbons. On the other hand, long range arity is defined by isolating residues separated by a sequence distance of $|i - j| > k$ that falls within a spatial radius of $d \leq r$.

The sequences of arity values are saved as independent and synchronized 1D spatial time series arrays, which are parameterized across multiple pairs of sequence separation k and spatial radius r values.

2.2 Phase 2: Topological Signatures—Mapping—Local Filtration

This phase extracts geometric baselines by expanding the distribution metrics and introducing local topological smoothing to distinguish ordered from disordered structural regions.

To begin with, the **Normalized Arity Histogram** is computed and an expanded **Cumulative Distribution Function** suite is used. Let $L = [A_1, ..., A_m]$ be the unique arities sorted by increasing value. The C_α **Arity Signature** for a number of specific percentiles $P_K = \{10th, 25th, 50th, 75th, 90th\}$ is defined as the smallest arity such that the CDF is larger than p_k:

$$arity_C_\alpha(p_k) = CDF^{-1}_{C_\alpha}(p_k) \equiv \arg\min_j (CDF_{C_\alpha}(A_j) \geq p_k)$$

Additionally, hyperparameter configurations (r, k, W) are systematically evaluated by calculating the Euclidean distance between AF2 and PDB signatures in 2D and 3D space. The configuration minimizing the mean D_{sig} across the dataset optimally captures the general geometric accuracy.

Moreover, the 25th, 50th and 75th percentiles are plotted to create discrete **2D and 3D maps**. At this global level, proteins physically cluster based on their macroscopic packing density, which means that rigid globular proteins accumulate more in the high arity top right region of the maps, while IDPs accumulate in the low arity bottom left region.

Furthermore, a **centered moving average** filter is applied across the 1D sequence arrays, that utilizes a minimum period of 1 to bridge sequence gaps introduced by the parameter k. Varied window sizes, specifically $W \in \{5, 11, 15, 25, 31\}$ are utilized that directly map to physical protein hierarchy: $W = 5$: Captures local secondary structure, like α-helices and β-strands. $W = 11$ to 15: Targets super secondary motifs. $W = 25$ to 31: Acts as a heavy low pass filter ideal for isolating macrodomains.

Subsequently, the C_α atoms are sorted by increasing arity and the Union-Find algorithm is run to track connected components. Also, topological quantifications such as Persistence Entropy ($H_\mathfrak{p}$), Mean Persistence ($\overline{p}$), Fraction of Positive Critical points (f^+_{cp}), Maximum number of connected components (N^{max}_{cc}), and Persistent Local Maxima ($PLM(t_\nu)$) are extracted.

Finally, to **systematically evaluate** the vast hyperparameter space (r, k, W) without relying on visual inspection, the Euclidean Signature Distance D_{sig} between the AF2 and PDB distributions is calculated. For a 2D map, this is defined as $D_{sig} = \sqrt{(p_{25,PDB} - p_{25,AF2})^2 + (p_{75,PDB} - p_{75,AF2})^2}$. The parameter pair yielding the lowest mean D_{sig} across the dataset identifies where AF2 most accurately captures global packing density.

2.3 Phase 3: Structural Alignment—Quaternary Validation

This phase physically validates spatial arrangements using combined probability metrics and mathematical sequence alignments.

To begin with, the 1D Earth Mover's Distance (EMD) is used to compare the full Cumulative Distribution Functions between AF2 and PDB models:

$$EMD(PDB, AF2) = \int_0^\infty |CDF_{PDB}(x) - CDF_{AF2}(x)|\, dx$$

Additionally, Dynamic Time Warping is deployed on the smoothed topology arrays from Phase 2. In order to handle unresolved PDB coordinates without distorting structural topology, the missing sequence parts are uniformly removed from both the experimental and predicted arrays before the DTW alignment. The cost matrix incorporates a weighting factor driven by AlphaFold's per residue

confidence pLDDT to secure alignment on highly structured domains, which allows low confidence regions to be bypassed with minimal penalty:

$$Cost(i,j) = \left(\frac{pLDDT_{AF2,j}}{100}\right)(Arity_{PDB,i} - Arity_{AF2,j})^2$$

Moreover, a structural hallucination flag is triggered if there is a highly confident and well folded local domain structure matching experimental reality but it is trapped in a wrong quaternary domain orientation:

$$(\text{DTW Error} \leq 1.0) \wedge (\text{Cross-Domain PAE} > 15.0\ \text{Å}) \wedge (\text{Mean Domain pLDDT} \geq 70.0)$$

The above condition ensures intrinsically disordered regions with naturally high DTW error and low confidence are excluded from false positive tracking.

Conclusively, to isolate true quaternary uncertainty, structural hallucinations are first flagged by applying a minimum sequence separation mask of 50 residues to the PAE matrix, which represents the established biophysical minimum size of an independently folding domain. With this mask, flagged structural hallucinations are validated by calculating the rigid body angle between the centers of mass of the distinct domains. To do this 1D signal processing is deployed on the inverted smoothed arity sequence to locate the precise topological saddle point and by enforcing a Topological Prominence threshold of 2.0 this continuous 1D structural minimum perfectly mirrors the discrete Persistent Local Maxima threshold ($T_\nu = 2.0$) from Phase 2, which means that at this specific point in the chain the packing density drops by at least 2 full neighbors compared to the solid domains on either side of it. As a result, the physical hinge coordinate H to compute the macroscopic fold angle θ is isolated:

$$\theta = \arccos\left(\frac{\vec{v}_A \cdot \vec{v}_B}{|\vec{v}_A||\vec{v}_B|}\right)$$

2.4 Phase 4: Statistical Benchmarking — Structural Ablation

The final phase statistically validates AlphaFold's topological deviations from experimental reality with multiple methods.

There are two methods used in this project to create clusters defined by cumulative arity density to ensure consistent biological labeling and also compare which configurations produce the optimal results.

Firstly, to capture the continuous evolution gradient of protein pack density, the ground truth biological modes are established by applying Agglomerative Hierarchical Clustering with $N = 4$ number of modes, to the baseline experimental PDB Arity Signatures (p_{25}, p_{75}). This isolates proteins into continuous structural families, which are specifically True IDPs, Molten Globules, Dynamic Globulars, and Rigid Globulars. These baseline assignments are held fixed across all hyperparameter configurations to serve as a stable scientific control group.

Secondly, to systematically isolate AlphaFold's structural biases, Agglomerative Clustering is again utilised but now executed independently on both the PDB and AF2 arity signatures for each configuration. The resulting mode assignments are fed into a confusion matrix. This quantifies the exact percentage of proteins that migrate across structural states when the k configurations change, like IDPs incorrectly predicted as Molten Globules, and hence reveals the most precise configuration with the most identical modes between PDB and AF2.

Moving forward, the filtration metrics (Persistence Entropy, Mean Persistence, Persistent Local Maxima) and Percentile distributions are grouped into paired sets of AF2 vs. PDB. Again non parametric Wilcoxon signed rank paired tests can be applied, by pairing every p-value with an Effect Size metric like Cliff's Delta to systematically quantify where AlphaFold's internal topological wiring deviates from experimental ground truth.

Additionally, the subset of proteins flagged as hallucinations in Phase 3 is isolated. The central tendency and extremes of their macroscopic rigid body angle deviations are quantified to spot the severity of AlphaFold's quaternary misalignments.

Finally, it is systematically tested whether the addition of multiscale long range arity yields statistically superior structural alignment compared to standard local arity. Using a paired Wilcoxon signed-rank test on the DTW alignment and EMD distributions, predictions processed with standard arity $k = 0$ are compared against those processed with long range isolating gaps $k > 0$, and specifically $k = 5$, to quantify the magnitude of information gain.

3 Results & Statistical Evaluation

Following the processing of $N = 8,050$ paired AlphaFold 2 and PDB experimental structures, the pipeline systematically evaluates all the above across the hyperparameter grid.

3.1 Baseline Evaluation

Operating at standard local arity ($r = 10$ Å, $k = 0, W = 1$), the baseline analysis in Table 1 reveals a systemic discrepancy in predictive accuracy between structural modes. As expected, AlphaFold aligns highly rigid structures in Mode 3 with better accuracy with a DTW of 2.07 compared to the rest of the modes. This geometric bias is also mirrored in the EMD, where the global distribution error steadily collapses as structural rigidity increases, dropping from 1.62 for True IDPs down to 0.49 for Rigid Globulars. Furthermore, the baseline data indicates that AlphaFold natively generates substantial topological noise, outputting 64.0 Persistent Local Maxima compared to the PDB's 30.0, which suggests the hallucination of microscopic false positive sequences.

Category	**Metric Evaluated**	**Values (AF2 vs PDB)**
Parameters	Radius (r), Gap (k), Window (W)	$r = 10, k = 0, W = 1$
	Total Proteins Analyzed	8,050
Alignment Error by Structure Mode	**Median DTW (Modes 0 / 1 / 2 / 3)**	2.58 / 2.65 / 2.76 / 2.07
	Median EMD (Modes 0 / 1 / 2 / 3)	1.62 / 1.12 / 0.82 / 0.49
	Mean 2D Signature Distance (D_{sig})	4.04
Topological Noise	Median Max Connected Components (CC)	50.0 (AF2) vs 27.0 (PDB)
	Persistent Local Maxima (PLM)	64.0 (AF2) vs 30.0 (PDB)
Topological Lifespan	Mean Persistence	5.85 (AF2) vs 6.48 (PDB)
	Persistence Entropy ($H_{\mathfrak{p}}$)	0.924 (AF2) vs 0.921 (PDB)
	Fraction of Pos. Critical Points (f_{cp}^{+})	0.228 (AF2) vs 0.249 (PDB)
Packing	25th / 50th / 75th Percentile Arity	AF2 (10 | 14 | 19) vs PDB (12 | 16 | 20)
Quaternary	Total Hallucinations Detected	159
	Mean Hallucination Angle Deviation	10.38°
	Max Hallucination Angle Deviation	120.56°

Table 1: Baseline Metrics. Modes represent the density gradient: True IDPs (0) → Rigid Globulars (3).

Biophysical Category	Metric Evaluated	p-value	Cliff's Delta (d)
Packing Density Gradient	90th Percentile Arity (p_{90})	5.38×10^{-10}	-0.012
	75th Percentile Arity (p_{75})	8.53×10^{-122}	-0.110
	50th Percentile Arity (p_{50})	< 0.001	-0.254
	25th Percentile Arity (p_{25})	< 0.001	-0.418
	10th Percentile Arity (p_{10})	< 0.001	-0.537
Topological Metrics	Max Connected Components (CC)	< 0.001	+0.479
	Persistent Local Maxima (PLM)	< 0.001	+0.440
	Mean Persistence ($\overline{p}$)	< 0.001	-0.313
	Fraction Pos. Critical Points (f_{cp}^{+})	< 0.001	-0.363
	Persistence Entropy ($H_{\mathfrak{p}}$)	5.67×10^{-11}	+0.058

Table 2: Baseline statistical insights evaluated via paired Wilcoxon signed-rank tests to calculate the exact continuous deviation of AF2 from experimental ground truth.

Regarding Table 2, due to the massive sample size standard p-values approached zero, therefore Cliff's Delta d was utilized to quantify the true magnitude of AlphaFold's structural deviations. The continuous packing differences are profound because while AlphaFold models the dense hydrophobic core p_{90} almost perfectly with a $d = -0.012$, its packing density progressively loosens toward the surface. Complementary, in Table 1, it is evident that AlphaFold strips away two full spatial neighbors from the median global fold as shown in the 50th percentile Arity with AF2's $p_{50} = 14$ vs PDB's 16, a reduction found also in the other percentiles. Simultaneously, AlphaFold exhibits underpacking of loose surface loops at the 10th percentile and 25th percentiles, as we can see from their respective Cliff's Delta values, which mathematically proves that AlphaFold renders protein surfaces far more expanded than they actually are. When looking at its topology, AlphaFold shows a lot of internal fragmentation with a CC metric at $d = +0.479$, while it also creates tiny false positive loops compared to experimental data, giving it a PLM d value of $+0.440$. Additionally, the topological features it predicts have noticeably shorter lifespans, shown by the variable $\overline{p}$ where $d = -0.313$, and it also consistently forms fewer positive critical loops as well, which is measured by f_{cp}^{+} at $d = -0.363$. All of this happens even though there is a slight increase in the overall persistence complexity, marked as $H_{\mathfrak{p}}$ with $d = +0.058$. Conclusively, this baseline pipeline flags 159 instances of quaternary hallucinations which have structurally sound local domains that are arranged with a mean rigid body deviation of 10.38° from experimental reality.

3.2 Long Range Geometry Evaluation

Evaluating the global dataset at the $k = 5$ sequence gap triggers important structural shifts, quantified in Table 3.

By enforcing a sequence gap of $k = 5$, α-helices and β-strands are mathematically deleted from the neighborhood counts, which triggers a massive alignment rescue for Intrinsically Disordered Proteins. The median DTW error for Mode 0 plummets from 2.58 at baseline down to 0.49, while the EMD drops from 1.62 to 0.50. This mathematically confirms that AlphaFold's primary geometric failure in IDPs is the hallucination of local secondary structures, rather than macroscopic misplacement. Consequently, the global 2D Signature Distance D_{sig} improves by a margin of 0.43 in the whole dataset.

Furthermore, removing this local noise causes the absolute density to drop a lot as it can be observed, where the 25th percentile plummets from 10 at baseline down to just 1 for AF2, compared to the PDB's 4, which means that the entire physical distribution shifts toward the origin in the 2D maps, leaving only the sparse and long range folding network visible. As summarized in Table 4, looking at these isolated tertiary networks reveals that the difference in topological lifespan gets a lot smaller, with $\overline{p}$ at $d = -0.110$. However, the problem with the packing gradient remains, as AlphaFold 2 keeps almost

Category	Metric Evaluated	Values (AF2 vs PDB)
Parameters	Radius (r), Gap (k), Window (W)	$r = 10, k = 5, W = 1$
	Total Proteins Analyzed	8,050
Alignment Error by Structure Mode	**Median DTW (Modes 0 / 1 / 2 / 3)**	0.49 / 1.46 / 2.12 / 1.64
	Median EMD (Modes 0 / 1 / 2 / 3)	0.50 / 0.71 / 0.63 / 0.40
	Mean 2D Signature Distance (D_{sig})	3.61 **(–0.43 shift)**
Topological Noise	Median Max Connected Components (CC)	46.0 (AF2) vs 25.0 (PDB)
	Persistent Local Maxima (PLM)	59.0 (AF2) vs 28.0 (PDB)
Topological Lifespan	Mean Persistence	6.04 (AF2) vs 6.26 (PDB)
	Persistence Entropy ($H_{\mathfrak{p}}$)	0.928 (AF2) vs 0.921 (PDB)
	Fraction of Pos. Critical Points (f_{cp}^{+})	0.205 (AF2) vs 0.240 (PDB)
Packing	25th / 50th / 75th Percentile Arity	AF2 (1 | 6 | 11) vs PDB (4 | 8 | 12)
Quaternary	Total Hallucinations Detected	716
	Mean Hallucination Angle Deviation	12.66°
	Max Hallucination Angle Deviation	145.05°

Table 3: Long Range Statistical Metrics with $k = 5$.

Biophysical Category	Metric Evaluated	p-value	Cliff's Delta (d)
Packing Density Gradient	90th Percentile Arity (p_{90})	7.45×10^{-28}	-0.028
	75th Percentile Arity (p_{75})	5.13×10^{-178}	-0.127
	50th Percentile Arity (p_{50})	< 0.001	-0.275
	25th Percentile Arity (p_{25})	< 0.001	-0.457
	10th Percentile Arity (p_{10})	< 0.001	-0.552
Topological Metrics	Max Connected Components (CC)	< 0.001	+0.442
	Persistent Local Maxima (PLM)	< 0.001	+0.422
	Mean Persistence ($\overline{p}$)	1.84×10^{-35}	-0.110
	Fraction Pos. Critical Points (f_{cp}^{+})	< 0.001	-0.465
	Persistence Entropy ($H_{\mathfrak{p}}$)	9.13×10^{-56}	+0.128

Table 4: Long Range Statistical Insights.

perfect tertiary packing in its core, marked by p_{90} at $d = -0.028$, but it still shows a huge errors in its smaller percentiles, where p_{10} is at $d = -0.552$.

From a topological standpoint, the general topology does not improve except the Mean Percistence error that gets smaller, which means that the individual long range loops AF2 predicts have structural prominences that closely match experimental reality. However, the overall arrangement of these features is deeply flawed, as the deviation in Persistence Entropy $H_{\mathfrak{p}}$ actually worsens and doubles from baseline to $d = +0.128$, therefore this mathematical shift suggests that AlphaFold's macroscopic long range networks are fundamentally messier and more uniformly tangled than the organised hierarchical domains seen in true experimental structures. This is further supported by the fact that the packing gradient flaw persists with the p_{10} d value be -0.552, while macroscopic domains remain highly fragmented with CC's d being +0.442.

It is also important to point out that the amount of Quaternary Hallucinations jumps from 159 to 716 when using the $k = 5$ setup, which is probably a mathematical result of having a sparser topology and the reduction of the mean DTW error, which enables the hallucination flagging to be triggered more frequently, so it would possibly be more ideal in future work to reduce this DTW threshold proportionally to the reduction of the mean DTW error in a certain configuration.

3.3 Windows Moving Average Reverberations

To statistically validate this multiscale approach, an ablation study was conducted (Table 5). It compares the baseline standard arity $k = 0$ against a minimal sequence gap $k = 1$ and a long range sequence gap $k = 5$. Additionally, this ablation was tested across four different sliding window setups, where the $W = 1$ setup means no window was used at all and for the other $W = 5$, $W = 15$, and $W = 25$ setups, the higher the number, the heavier the smoothing effect becomes. In fact, these different sizes actually correspond to smoothing out specific types of structures, specifically secondary motifs for $W = 5$, super secondary folds for $W = 15$, and macroscopic domains for $W = 25$. This was done to determine how spatial gaps k interact with frequency low pass filters W.

When looking at how things work in table 5, in all window cases just skipping the closest covalent neighbor, where $k = 1$, barely improves the alignment, as it can even slightly worsen the geometric signature distance, with D_{sig} effect sizes going from +0.001 to +0.002. On the other hand, skipping the whole secondary structural turn, where $k = 5$, causes a very strong and consistent drop in all errors across the entire dataset for all the parameters we checked and for all windows.

But there is a noticeable biophysical caveat when one looks at how the spatial gap k interacts with the sequence smoothing window W. At high resolution levels that make physical sense, like when $W = 1$ and $W = 5$, the sequence gap works as a precise signal rescuer and this gives DTW d values of -0.271 and -0.255. With these tighter windows, the pipeline does a good job separating the larger structural differences without losing the specific chemical details of the polypeptide chain in that local area.

In contrast, when the moving average window grows to larger sizes like $W = 15$ and $W = 25$, the specific effect of the $k = 5$ gap weakens a lot, as the DTW d drops down to -0.128 and -0.077. This drop does not mean the structures are actually matching up better, but instead it points to a clear limit where loss of information begins, because at a larger macrodomain window size like $W = 25$, the low pass filter aggressively flattens out the topological changes. While this mathematically smooths out the alignment differences, it also wipes out the specific local energy changes and unique thermodynamic traits that make a specific fold what it is. Because of this, even though massive smoothing windows are still useful for checking the outer boundaries of where global domains sit, the cases where they should be used must be really thought through, as excessive values will completely destroy the unique thermodynamic fingerprint of the natural protein structure.

Scale	Metric	Minimal Gap ($k = 0 \to 1$)		Tertiary Gap ($k = 0 \to 5$)	
		p-value	Effect Size (d)	p-value	Effect Size (d)
Raw Atoms ($W = 1$)	**DTW Alignment**	0.00	-0.035	0.00	**-0.271**
	EMD Global Score	0.00	-0.018	0.00	**-0.221**
	2D Signature Dist.	4.68×10^{-1}	+0.001	7.87×10^{-86}	**-0.053**
	3D Signature Dist.	5.70×10^{-1}	+0.001	7.67×10^{-53}	**-0.038**
Micro/Secondary ($W = 5$)	**DTW Alignment**	0.00	-0.047	0.00	**-0.255**
	EMD Global Score	0.00	-0.030	0.00	**-0.258**
	2D Signature Dist.	3.48×10^{-11}	+0.001	1.32×10^{-101}	**-0.047**
	3D Signature Dist.	1.07×10^{-12}	+0.001	1.36×10^{-73}	**-0.036**
Super-Secondary ($W = 15$)	**DTW Alignment**	0.00	-0.025	2.33×10^{-240}	**-0.128**
	EMD Global Score	0.00	-0.018	0.00	**-0.173**
	2D Signature Dist.	1.36×10^{-65}	+0.002	9.57×10^{-120}	**-0.048**
	3D Signature Dist.	1.56×10^{-61}	+0.001	2.76×10^{-88}	**-0.036**
Macro/Domain ($W = 25$)	**DTW Alignment**	1.74×10^{-298}	-0.014	1.20×10^{-110}	**-0.077**
	EMD Global Score	1.32×10^{-147}	-0.009	4.08×10^{-287}	**-0.121**
	2D Signature Dist.	7.08×10^{-78}	+0.001	1.08×10^{-131}	**-0.050**
	3D Signature Dist.	1.06×10^{-68}	+0.001	1.33×10^{-105}	**-0.041**

Table 5: Comprehensive Sequence gap Ablation Study across multiple smoothing windows. A larger negative Cliff's Delta d indicates a stronger reduction in predictive error against the PDB ground truth.

3.3.1 Optimal Hyperparameters for Mean D_{sig} Error

To assess the hypothesis that isolating long range arity improves structural alignment by removing trivial secondary backbone contacts, a global grid search was executed over the hyperparameter space and one optimal configuration was selected according to the mean D_{sig} error.

Configuration Type	Parameters	Mean D_{sig} Error
Experimental Baseline	$r = 10$ Å$, k = 0, W = 1$	4.044
Raw Tertiary Shift	$r = 10$ Å$, k = 5, W = 1$	3.610
Biophysical Optimum	$r = 10$ Å$, k = 5, W = 5$	3.465
Mathematical Asymptote	$r = 10$ Å$, k = 5, W = 31$	3.420

Table 6: Optimal Configuration for Mean D_{sig} Error

While a purely algorithmic check shows that the heavy low pass filter where $W = 31$ gives the absolute mathematical lowest point for global topological error, with $D_{sig} = 3.420$, the biophysical reality limits how useful it actually is, because massive smoothing windows cause a loss of information that flattens out the specific local energy variations of the natural fold, as already stated before. As a result, the sweet spot is probably at $W = 5$ and $k = 5$. With this setup, the pipeline successfully gets past the local secondary noise to focus on the tertiary backbone, dropping the D_{sig} to 3.465. Selecting also the $W = 1$ and $k = 5$ configuration remains valid because the reduction from the baseline is still quite significant at around 0.4 . Expanding the smoothing window all the way to $W = 31$ only gives a mathematically tiny alignment improvement of just 0.045 from the $W = 5$ configuration. This proves

that $W = 5$ catches the vast majority of the real structural signal without crossing into the zone where important information is lost.

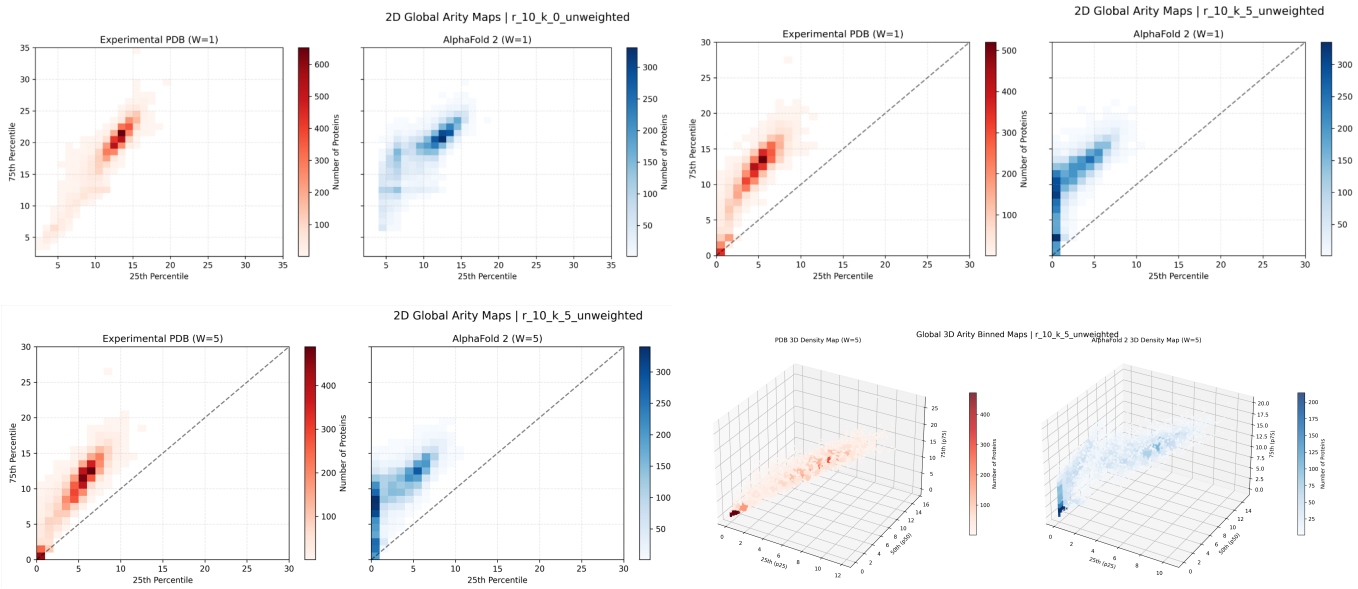


Figure 1: Top Left: $k = 0, W = 1$. Top Right: $k = 5, W = 1$. Bottom: $k = 5, W = 5$ in 2D and 3D.

3.4 Hierarchical Clustering Modes — Taxonomic Agreement

To extract biological meaning from these geometric shifts, Agglomerative Hierarchical Clustering partitioned the PDB dataset into a continuous gradient ranging from IDPs to Rigid Globular folds, as shown in previous tables. Table 7 quantifies the alignment improvement generated by the $k = 5$ filter across these distinct evolutionary states.

Structural Mode	**Dataset Size (N)**	**$k = 0$ DTW / EMD**	**$k = 5$ DTW / EMD**	**Improvement**
Mode 0: True IDPs	664	2.58 / 1.63	0.49 / 0.50	**-2.09 / -1.13**
Mode 1: Molten Globules	929	2.65 / 1.12	1.46 / 0.71	**-1.19 / -0.41**
Mode 2: Dynamic Globulars	3,556	2.76 / 0.82	2.12 / 0.63	**-0.64 / -0.19**
Mode 3: Rigid Globulars	2,901	2.07 / 0.49	1.64 / 0.40	**-0.43 / -0.09**

Table 7: Median DTW and EMD Error reduction across the evolutionary packing density gradient.

This mode by mode breakdown reveals the core signal to noise dynamics of AlphaFold. **Mode 0 IDPs** experience a massive rescue with a -2.09 DTW reduction because AlphaFold frequently hallucinates transient local secondary structures in floppy regions, as it was already stated. Setting $k = 5$ explicitly blinds the evaluation algorithm to these local hallucinations, forcing a comparison strictly on long range architecture, which AlphaFold identifies more precisely.

Following this topological trend, **Mode 1 Molten Globules** experience an intermediate rescue with a -1.19 DTW reduction. Because they possess secondary motifs but lack a completely rigid core, the $k = 5$ filter successfully removes their local noise, though AF2's persistent struggle with their loose macroscopic packing limits further improvement.

Conversely, **Mode 2 Dynamic Globulars** see only a moderate rescue with a -0.64 DTW reduction and retain the highest DTW errors in both k cases. While AlphaFold correctly predicts their local secondary helices, it frequently fails at predicting their dynamic tertiary structures, opting to overpack them into single rigid states. Setting $k = 5$ probably strips away the secondary structures AF2 got right, isolating and exposing its inherent tertiary packing flaws. These result to Mode 2 having the highest DTW error of all other modes.

Finally, **Mode 3 Rigid Globulars** exhibit the smallest alignment rescue with a -0.43 DTW reduction, but also having lower DTW error than Mode 2 and the lowest EMD error of all the other modes in both k cases. Because these proteins are inherently highly structured, AlphaFold's baseline predictions of their dense local geometries are already highly accurate. The residual error exposed at $k = 5$ is almost entirely driven by macroscopic surface expansion, rather than local secondary misalignment that AlphaFold predictions generally produce.

Confusion Matrix at Baseline $(r = 10, k = 0, W = 1)$ **– Accuracy: 42.87%**				
	Pred AF2 0	**Pred AF2 1**	**Pred AF2 2**	**Pred AF2 3**
True PDB 0 (IDP)	310	217	101	36
True PDB 1	391	438	83	17
True PDB 2	381	1551	1319	305
True PDB 3 (Rigid)	85	500	932	1384

Table 8: Baseline Taxonomic Agreement PDB Ground Truth vs. AF2 Prediction.

Confusion Matrix at Long-Range Topology $(r = 10, k = 5, W = 1)$ **– Accuracy: 49.66%**				
	Pred AF2 0	**Pred AF2 1**	**Pred AF2 2**	**Pred AF2 3**
True PDB 0 (IDP)	849	360	241	75
True PDB 1	336	775	854	162
True PDB 2	176	423	1446	1020
True PDB 3 (Rigid)	31	32	342	928

Table 9: $k = 5$ Taxonomic Agreement PDB Ground Truth vs. AF2 Prediction.

Best Taxonomic Configuration Reference $(r = 10, k = 0, W = 11)$ **– Accuracy: 56.05%**				
	Pred AF2 0	**Pred AF2 1**	**Pred AF2 2**	**Pred AF2 3**
True PDB 0 (IDP)	197	188	116	36
True PDB 1	362	685	257	24
True PDB 2	200	1279	2423	336
True PDB 3 (Rigid)	21	201	518	1207

Table 10: Best Taxonomic Agreement Configuration.

Now in this part Agglomerative Clustering is again utilised but executed independently on both the PDB and AF2 arity signatures for each configuration to produce confusion matrices that contain the experimentally true and misclassified numbers of proteins for each mode. Specifically, at the baseline (r = 10, k = 0, W = 1) AlphaFold systematically overpacks IDPs, because out of 664 experimentally confirmed IDPs in row 0, AlphaFold misclassifies 217 + 101 + 36 = 354 proteins into denser modes, mathematically confirming the inflation of local density via structural hallucinations. Conversely, AlphaFold significantly underpacks rigid globulars, as out of 2,901 True PDB Rigid Globulars in row 3, AF2 misclassifies exactly 1,517 of them into looser modes. This exposes the **Crystal Lattice effect**, because experimental PDB structures are compressed via X-ray crystallography while AlphaFold predicts proteins freely in solution so the predicted rigid cores are naturally rendered slightly more

expanded than their crystal counterparts. This can be also easily observed in Mode 2, where a lot of Dynamic Globulars are misclassified for less rigid modes, like Mode 1, more often than more rigid modes.

Increasing the sequence gap to $k = 5$ physically isolates the true long range packing, elevating the overall taxonomic agreement from 42.87% to 49.66%, but also changes decisively the number of proteins each mode has and produces really different modes than the baseline. However, exhaustive grid search evaluation reveals an inverse correlation between spatial gaps k and moving average smoothing W. The absolute highest taxonomic accuracy 56.05% is achieved by leaving standard arity intact and by applying a super secondary smoothing window $W = 11$. This indicates that topological noise can be filtered via k sequence gaps or moving average smoothing, but deploying both filters simultaneously reduces taxonomic accuracy, something that was actually shown in every case. For example the configuration $k = 5, W = 11$ drops accuracy to 44.41%.

Ultimately, this $W = 11$ configuration is the maximum boundary for taxonomic classification, but even the $k = 5$ sequence gap filtering gives an increase in taxonomic accuracy without smoothing the protein sequences and losing biophysical credibility, like $W = 11$ does. Logically, pushing the smoothing window further with $W \geq 15$ causes the accuracy to systematically destabilize, confirming again that excessive low pass filtering forcibly blends the topological signatures of distinct evolutionary states and severely degrades the methodology's biological credibility.

4 Discussion

This study implemented a topological pipeline to assess the structural predictions of AlphaFold 2 against experimental PDB data, by evolving the baseline Arity metric to incorporate long range sequence gaps k and moving average low pass filters W. The integration of geometric signature distances D_{sig}, Dynamic Time Warping and Earth Mover's Distance yields theoretical and practical outcomes regarding AlphaFold's predictive capabilities and internal biophysical biases across 8,050 paired protein structures.

The comprehensive findings of this study yield the following definitive conclusions:

- **IDP Hallucinations & Local Artificial Densification:** AlphaFold systematically struggles with IDPs and hyper-compacts them into denser structural modes. The alignment rescue observed when introducing the $k = 5$ sequence gap mathematically proves that AF2's primary failure in flexible regions is the hallucination of transient local secondary structures. Blinding the metric to this local noise reveals that AF2 actually recognizes the macroscopic emptiness of IDPs quite well.

- **Sequence Gaps Flaws:** AlphaFold is good at predicting the local secondary helices of both Dynamic Globulars and Molten Globules, but it fails to capture how these proteins naturally move in space, because when the correct secondary structures are stripped away by using the $k = 5$ gap, AlphaFold's deeper packing flaws become obvious. Its large scale and long range networks are highly fragmented with elevated Connected Components and are also topologically tangled, which is shown by a worse Persistence Entropy compared to naturally organized protein domains.

- **The Crystal Lattice Effect & Surface Porosity:** On the other hand, AF2 systematically loosens Rigid Globulars compared to their experimental counterparts. When evaluating the continuous packing density gradient it was revealed that while AF2 predicts the dense hydrophobic core p_{90} almost perfectly, it severely expands loose surface domains in p_{10}. This is probably a reflection of the Crystal Lattice Effect, where AF2 predicts the natural and slightly expanded state of the protein in solution and bypasses the artificial compression, which is a characteristic of X-ray crystallography. So this is probably not a problem of the Alphafold 2 model rather than a drawback of the crystallized experimental dataset.

- **The Moving Average Information Decay:** While heavy low pass filters mathematically minimize global D_{sig} signature errors, they trigger severe biophysical information decay. The study identifies this problem and proposes $W = 5$ as a maximum window value for isolating tertiary architecture without destroying the local variations that define a fold's unique thermodynamic fingerprint. Pushing smoothing windows higher forcibly blends distinct evolutionary states and degrades biological credibility.

- **The k vs. W Inverse Correlation:** The methodology proves that spatial gaps k and moving average smoothing W windows are mutually exclusive filters for noise reduction. Yet, applying both simultaneously, a configuration like $k = 5, W = 11$, severely filters the arrays to the point of worsening taxonomic accuracy. The absolute highest taxonomic agreement 56.05% is achieved using standard arity $k = 0$ with a moving average filter of $W = 11$.

- **Quaternary Hallucination Sensitivity:** The pipeline successfully flagged structurally sound local domains arranged with severe quaternary rigid body angle deviations. However, isolating the sparser long range topology $k = 5$ caused detection rates to spike from 159 to 716 instances. This outcome highlights that when the backbone noise is removed, dynamic adjustments to the DTW inclusion thresholds are required in future applications to prevent excessive flagging.

5 Conclusion

Ultimately, this multiscale analysis highlights a fundamental bias within AlphaFold 2, which is the fact that there is a structural regression to the mean value. Because AlphaFold is a deep learning model trained on standard, stable and well-folded globular proteins, its learning parameters are strongly biased toward this middle ground. Especially, when presented with the extreme flexibility of an IDP, the algorithm forces it to resemble a normally folded protein, resulting in the artificial densification and hallucination of tight local helices. On the other hand, when predicting a rigid globular protein that is natively squeezed tightly within an X-ray crystal lattice, AlphaFold predicts it as it would exist in solution, causing highly compressed surface loops to relax and become overly porous. AlphaFold is not randomly committing two opposite geometric errors, it is rather systematically pulling the two extremes of protein biology, which are the overly floppy and the highly rigid extremes, toward the average globular state in solution upon which it was predominantly trained. The $k = 5$ sequence gap is particularly successful in mitigating the IDP extreme.

While the optimal configurations identified in this study successfully balance sequence gaps and frequency filters to expose these thermodynamic realities, this methodology must be applied with biophysical intuition. As demonstrated, deploying excessive sliding windows destroys the biophysical semantics of the protein array. Furthermore, the hyperparameter grid search conducted in this study was bounded by available processing constraints, so leveraging more extensive computational power could evaluate a vastly expanded (r, k, W) matrix, potentially yielding even better configurations and further advancing the biophysical understanding of AlphaFold.

Future Work

Because the PDB is inherently biased toward rigid and crystallizeable proteins, future methodological expansions should replace the PDB baseline with targets curated from *DisProt* for validated IDPs or the *Protein Ensemble Database*. Validating AlphaFold against NMR structural ensembles rather than static X-ray, would allow this topological pipeline to test AF2's geometries against true thermodynamic realities. Additionally, implementing distance decay exponential weights in Phase 1 could capture more nuanced atomic packing gradients.

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